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# Research and application of semantic understanding based on Attention-RNN

Wen-yong DU<sup>a</sup>, Li ZHANG<sup>a</sup>, Lin-tan SUN<sup>a</sup>, Yu-hang CHEN<sup>b,\*</sup>, Cheng-huan LI<sup>a</sup>

<sup>a</sup>Customer Service Center of SGCC/Tianjin City/China

<sup>b</sup>Information System Integration Company, NARI Group Corporation, Nanjing City, Jiangsu Province, China

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## Abstract

Spoken language understanding module is an important part of the human-machine dialogue system, which contains two important subtasks of intent recognition and slot filling. In the current human-machine dialogue system, intent recognition and slot filling are mostly in the form of independent tasks. Exist, it is difficult to further mine the latent semantic information in the text. This paper attempts to introduce attention mechanism in the process of intent recognition, and share the latent information of intent recognition and slot filling to improve the result of intent recognition and slot filling. Experiments show that the introduction of attention mechanism and the sharing of hidden information are important for slot filling and intent recognition and have a certain improvement effect.

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**Keywords:** Natural language understanding; deep learning; attention mechanism; intent recognition; slot filling; dialogue system; semantic understanding;

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## 1. Introduction

In recent years, with the rapid iterative update of computer hardware and the rapid growth of computing power, deep learning represented by deep neural networks has ushered in the third wave, and has achieved fruitful results in speech recognition, image recognition and other fields<sup>[1]</sup>. Thanks to the excellent performance and characteristics of the deep model, researchers try to introduce deep learning into the existing semantic understanding framework to build an end-to-end semantic understanding and slot filling model. However, in traditional engineering practice,

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\* Corresponding E-mail address: [chenyuhang1@sgepri.sgcc.com.cn](mailto:chenyuhang1@sgepri.sgcc.com.cn)

intent recognition and slot filling are mostly the research methods of separate modeling, which artificially separates the correlation between the two tasks<sup>[2]</sup>. Therefore, more and more researchers adopt the research method of joint modeling and use the correlation information between intent recognition and slot filling to simplify the complexity of semantic understanding to a certain extent.

This paper proposes a semantic understanding model based on Attention-RNN. This model is based on the encoding-decoding structure, and uses the idea of joint modeling to cascade the intention recognition model and the slot filling model to construct an end-to-end semantic understanding model<sup>[3]</sup>, using the intention and slot relevant information to improve semantic understanding results. The model introduces an attention mechanism during decoding, captures semantic information according to the importance and relevance of each position in the text, improves the accuracy of slot filling and intent recognition, and enhances the accuracy of the semantic understanding model on long text sequences.

## 2. Semantic understanding model based on Attention-RNN

In traditional semantic understanding tasks, modularization is usually used to model the two tasks of intent recognition and slot filling separately to obtain two semantic understanding components, and finally merge the results of these two components to complete Semantic understanding of user sessions<sup>[4]</sup>. However, in this process, because the two tasks are performed independently and the semantic understanding angle is single, it is difficult to comprehensively analyze the semantics of the entire conversation text from multiple dimensions. The intent recognition results and the slot filling recognition results are often difficult to match. Therefore, this article attempts to use the encoding and decoding structure of the Seq2Seq model as the basis to combine the two tasks of intent recognition and slot filling to establish an end-to-end semantic understanding model, so that the intent can be obtained at the same time with only one training. Recognition result and slot filling result. Through the sharing of the coding layer, in the training process, the intention of the original text and the hidden information of the slot influence and merge each other, so that the entire semantic understanding model can learn more and more comprehensive latent semantic information.

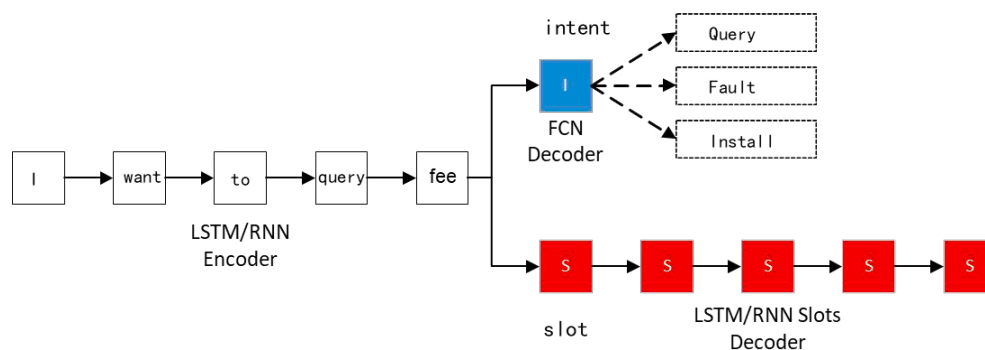


Fig. 1. Joint model of language understanding based on encoder-decoder structure

In the above model, the traditional approach is to use the output at the last time  $t$  of the coding layer as the  $n$ -dimensional semantic feature representation of the original input sequence, and then decode this vector to obtain the corresponding intent and slot information. But a potential problem of this method is that once the model tries to compress all the information of a long sequence of text into a fixed-length vector, it will inevitably lead to the loss of some information. At the same time, as the length of the text increases, this loss phenomenon becomes more obvious, and the performance of the model drops quickly.

In order to reduce the effect of information loss on the performance of the model, this paper tries to introduce an attention mechanism in the decoding stage to retain as much effective information as possible according to the inherent importance and relevance of the information. As shown in Figure 2, this article introduces attention blocks

for the intention decoding layer and the slot decoding layer, and uses the attention blocks to preserve important information in the original input sequence as much as possible.

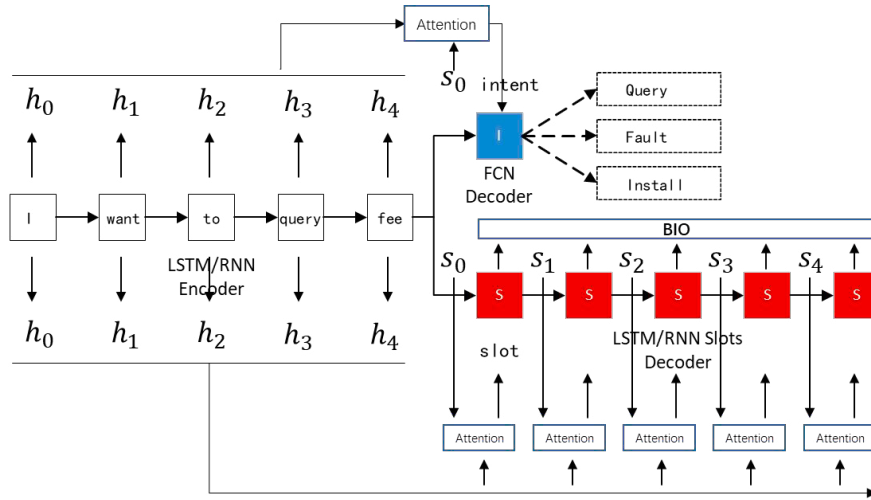


Fig. 2. Language understanding model with attention mechanism

### 3. Experiment

ATIS is an open source English data set frequently used in researches such as oral comprehension and intention recognition, and it is marked by BIO. The data set has a total of 945 words, including 4978 pieces of data in the training set and 893 pieces of data in the validation set, containing a total of 26 intents and 130 slot labels.

The ECSF data set is collected from the Chinese question and answer text between the manual customer service and the user in the electric bill field of the customer service center. There are 3260 pieces of data, including 10 intents and 13 slot tags, as shown in Table 1 and Table 2.

Table 1. Intents of ECSF dataset

| Intent Type                             | Example                                                            |
|-----------------------------------------|--------------------------------------------------------------------|
| Electricity fee inquiry                 | How much did the electricity bill last month                       |
| Electricity fee consultation            | How is the electricity price in Hedong District calculated?        |
| Electric meter query                    | Please help me look at the number of my electric meter             |
| Abnormal battery query                  | The battery has been used incorrectly recently, why is it so much? |
| Inconsistent amount query               | I charged 120 yuan, how come it only shows that 100 is received    |
| Repeat payment query                    | Oh, why did that system charge me another electricity bill         |
| Cancellation notice of electricity bill | Help me cancel that monthly electricity bill SMS                   |
| Not received electricity bill inquiry   | The bill from last month was not sent                              |
| Power outage query                      | Why is there a power outage in my community?                       |
| Business hall query                     | Where is the nearest business office nearby                        |

Table 2. Slots of ECSF dataset

| Slot Type | Example                                                      |
|-----------|--------------------------------------------------------------|
| User      | User name, user number, mobile phone number, identity number |
| Device    | Device name, device type, device color, device status        |
| Other     | Location, date and time, organization, quantity              |

In this paper, a joint modeling method is used to construct a semantic understanding model M2 based on the encoding and decoding structure, and then the attention mechanism is introduced in the decoding stage to obtain the semantic understanding model M3 based on the attention mechanism, and the two models are respectively in the ATIS data set and Test it on the ECSF data set and compare it with the results of the intent recognition and slot filling model M1 using separate modeling.

Table 3. Performance of different models

| Model | ATIS    |          | ECSF    |          |
|-------|---------|----------|---------|----------|
|       | F-score | Accuracy | F-score | Accuracy |
| M1    | 87.8    | 88.9     | 75.1    | 84.3     |
| M2    | 94.7    | 91.3     | 82.2    | 82.6     |
| M3    | 95.1    | 95.7     | 90.3    | 87.8     |

Table 3 shows the results of testing different models on the ATIS dataset and the ECSF dataset. Among them, M1 is the independent task semantic model, that is, intention recognition and slot filling are performed separately; M2 is the fusion model proposed in this paper, namely Intention recognition and slot filling share the coding layer; M3 is the fusion model proposed in this paper after introducing the attention mechanism. It can be seen that the fusion model M3 after introducing the attention mechanism has significantly improved intent recognition accuracy and slot filling accuracy compared to M1 and M2.

#### 4. Conclusion

This paper uses deep learning methods to integrate traditional intent recognition and slot filling tasks, and introduces an attention mechanism to improve model performance. Experiments show that the model proposed in this paper can effectively improve the performance of intention recognition and slot filling, and achieves better results on ATIS and ECSF.

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